

40. Sharky, a leader of the underworld, was killed by one of his own band of four henchmen. Detective Sharp interviewed the men and determined that all were lying except for one. He deduced who killed Sharky on the basis of the following statements:

- a. Socko: Lefty killed Sharky.
- b. Fats: Muscles didn't kill Sharky.
- c. Lefty: Muscles was shooting craps with Socko when Sharky was knocked off.
- d. Muscles: Lefty didn't kill Sharky.

Who did kill Sharky?

L = Lefty killed Sharky
M = Muscles killed Sharky
S = Socko killed Sharky
F = Fats killed Sharky

A = Muscles was shooting craps with Socko when Sharky was knocked off.

- a. L
- b. $\sim M$
- c. A
- d. $\sim L$

We know that the detective determined that they were all lying except for one.

Case 1: Socko is telling the truth

$L \wedge M \wedge \sim A \wedge L$

If Socko is telling the truth, then Lefty killed Sharky, and Muscles killed Sharky, and Muscles was not shooting craps with Socko when Sharky was knocked off, and Lefty killed Sharky.

This statement becomes contradictory, since we know Sharky was killed by only one of his associates, and if Socko was telling the truth, then that means both Lefty and Muscles killed Sharky.

Socko is therefore lying.

Case 2: Fats is telling the truth

$$\sim M \wedge \sim A \wedge L \wedge \sim L$$

If Fats is telling the truth, then Muscles did not kill Sharky, and Muscles was not shooting craps with Socko when Sharky was knocked off, and Lefty killed Sharky, and Lefty did not kill Sharky.

This statement also becomes contradictory, since it states that Lefty both did- and did not kill Sharky.

Fats is therefore lying.

Case 3: Lefty is telling the truth

$$A \wedge L \wedge \sim L \wedge M$$

If Lefty is telling the truth, then Muscles was shooting craps with Socko when Sharky was knocked off, and Lefty killed Sharky, and Lefty did not kill Sharky, and Muscles killed Sharky.

Again, this statement is contradictory, since Muscles was both shooting craps with Socko, he killed Sharky together with Lefty, but Lefty did not kill Sharky.

Lefty is therefore lying.

Case 4: Muscles is telling the truth

$$\sim L \wedge \sim L \wedge M \wedge \sim A$$

If Muscles is telling the truth, then Lefty did not kill Sharky, and Lefty did not kill Sharky, and Muscles killed Sharky, and Muscles was not shooting craps with Socko when Sharky was knocked off.

This is the only case where the conditions are met. This statement also implicates Muscles, even though he is telling the truth.

Solution: Muscles killed Sharky.